library. The Pow[27] library gives you an integrated authentication solution that works quite well. (This is my personal authentication library of choice.)

Phoenix Channels

Real-time features have become a staple of the modern tech stack. Real-time applications push changes from the backend to the frontend so the updated data is available to users without them having to take action. This creates a seamless experience that can make an application feel polished and snappy.

Phoenix Channels[28] provide an out-of-the-box way to quickly add real-time features to your application. Channels are backed by Elixir's process model and use WebSockets to communicate in soft real-time between the frontend and backend.

My other book, *Real-Time Phoenix [Bus20]*, covers close to everything about channels. They come with their own complexities, but Phoenix's implementation is truly top-notch.

Phoenix LiveView

Phoenix LiveView[29] takes real-time applications to the next level. LiveView is built on top of channels to provide a new real-time programming model.

In LiveView, the server manages all of the HTML and state of your application. The server pushes the HTML to the client and ensures it's updated as the state changes on the server. Essentially, LiveView completely replaces controllers and a lot of JavaScript.

One of the great things about LiveView is that Phoenix templates and components work there. If you learn how to write templates and business logic with Phoenix Controllers, you can take everything you've learned and move it over to LiveView!

Make sure to check out *Programming Phoenix LiveView [TD24]* by Bruce Tate and Sophie DeBenedetto. It's the best book to get started with LiveView!

Wrapping Up

Phoenix is one of the main libraries for Elixir application development. Phoenix is a bit like Rails in that it is the most prominent library in the ecosystem, but it has a significantly different objective than Rails does. Rails seeks to be

27. https://github.com/pow-auth/pow
28. https://hexdocs.pm/phoenix/channels.html
29. https://hexdocs.pm/phoenix_live_view/welcome.html